



NSF Engineering Research Center
for Computer Integrated Surgical
Systems and Technology



LABORATORY FOR
**Computational
Sensing + Robotics**
THE JOHNS HOPKINS UNIVERSITY

**WHITING
SCHOOL OF
ENGINEERING**
THE JOHNS HOPKINS UNIVERSITY

601.455/655 Computer-Integrated Surgery

Fall Semester 2025

<https://ciis.lcsr.jhu.edu/doku.php?id=courses:455-655:455-655>

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Organizational Information

- **Instructor**
 - Russell H. Taylor (rht@jhu.edu)
 - Guest Lectures
- **Class Place & Times**
 - Tuesdays and Thursdays 1:30-2:45
 - Class will meet in Hackerman B17
 - Section times: TBD
- **TA**
 - Bohua Wan (bwan2@jhu.edu)
 - Office hours and section times: See the course web site (link below)
- **Textbook**
 - Handout Material (on web site)
 - Recommended: *Computer-Integrated Surgery*: (Editors: Taylor, Lavallee, Burdea, Mosges), MIT Press, 1996
- **Office hours**
 - By appointment, but we will usually be available before or after class
 - Some additional office hours may be available from selected PhD student CA's
- **Web site**: <https://ciis.lcsr.jhu.edu/doku.php?id=courses:455-655:455-655>
 - Links to lecture notes and homework found here
- **Piazza Page**: <https://piazza.com/class/mdyvtys9f3n5v4>
 - Mostly used for class communications and Q&A
- **Canvas Page**: Follow link for EN.601.455.01.FA24
 - Non-public links; pointers to video recordings; etc.
- **Gradescope**: Students will submit assignments on Gradescope ([gradescope.com](https://www.gradescope.com))
- The course entry code is D3G5NK



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Overview of the CIS Course Sequence

- **601.455/655 (Fall)**
 - Goal is to provide overview of basic techniques & applications and to provide background for subsequent research
 - Lectures + homework + programming
 - Optional “project” is usually a report and plan for subsequent implementation project
- **601.456/496/656 (Spring)**
 - Combination advanced seminar + lectures + student team projects
 - Emphasis on student projects + project management + presentation skills



601.455/655 Course content

- Basic concepts of computer-integrated surgery
 - Image segmentation, registration, modeling
 - Robotics
 - Safety
 - Human-machine interfaces
- Application case studies
 - Lectures by clinicians & systems implementers
- Outcomes and economic analysis
- Other topics of interest
 - E.g., regulatory affairs



601.455 vs 601.655

- 601.455 is intended for upper-level undergraduates
- 601.655 is intended for graduate students
- The course content will be the same.
- Students are **strongly** urged to work in teams of 2, but they may work independently.



Other courses

- **530/601.856 – Medical Image Analysis Seminar**
 - Spring semester
 - Papers on selected topics in medical image analysis
 - Jointly led by Prof. Prince & Prof. Taylor
 - Students read a paper every week & review it in detail during weekly recitation
- **500.745 – LCSR Seminar**
 - Wednesday 12-1pm
 - Various topics related to CIS and robotics research
 - Distinguished outside speakers + JHU faculty members + grad students talking about their work



Other CIS education activities

- **Computer-Integrated Surgery Minor**
 - WSE undergrad degree
 - CIS “Minor” advisor selected from approved faculty
 - CIS Course Sequence is base
 - Additional courses from selected menu / consent of advisor
 - Tracks for robotics & imaging
 - See me for information



Guest lectures

- Generally, will have between several lectures from outside speakers, either to cover times when Prof. Taylor has unavoidable travel or to broaden course
- Speakers/topics will include
 - Surgeons
 - Radiologists
 - Radiation oncologists
 - Regulatory issues
 - Health economics
 - Entrepreneurs
 - Specific technical topics



601.455/655 Prerequisites

- No hard and fast rules, but ...
- Mathematical background
 - You will need to work with coordinate transformations and linear approximations
 - Calculus will be assumed
 - Linear algebra is highly recommended
 - Probability/statistics will be helpful
- Programming
 - No specific languages required. Homework can be handed in in any “usual” language (C, C++, MATLAB, Python) but needs to be well discussed and documented.
 - We will need to be able to run your programs.
 - Example handouts will be in pseudo-code, C++ and/or C
 - But the language is less an issue than basic concepts
 - Familiarity with basic data structures is important
 - Your lab partner choice is important. Pick complementary skills



Piazza Page

- Piazza page is: <https://piazza.com/jhu/fall2025/601455655>. Primarily, this will be used for announcements and to provide a means for students to ask questions of the instructor and the TA. We will be monitoring it fairly closely, but don't guarantee instantaneous response.
- Students enrolled in the class need to activate their Piazza accounts and check to see that they have access to the page. If you have problems, contact the TA.
- Here are a few of the uses that are legitimate for the Q/A section of the page:
 - Advertising for a lab partner
 - Asking general questions about concepts in the class (though you may be referred to the TA)
 - Asking for clarification on homework (to be answered **only** by TA or instructor)
 - Pointing out typographical errors or other problems in handouts
- Here are some things that are not proper:
 - Giving or seeking specific help or providing answers to homework assignments
 - Electronic collaboration of any kind
 - Uses unrelated to the course



Homework Submission

- You may work individually or in teams of two
 - Teams of two are **strongly** recommended
- We will be using GradeScope for homework and programming problem submission.
- But in the meantime, please **sign up** at
 - Link: <https://gradescope.com>
- Code: D3G5NK
 - If you work in a team of two, be sure to have **both** your names on what you hand in.
 - Only **1 submission** per group (if you are in a group of 2, please use the "group functionality" on Gradescope)



Grades and homework

- **Course grade is based on take home assignments plus in-class quizzes**
 - 4 **take home exams** (called "homework") done in teams of two
 - Three person teams not permitted
 - Single person teams allowed but strongly discouraged
 - 5 **Programming assignments** done in teams of two (PA#5 optional)
 - Possible in-class quizzes done individually
 - Grading formula discussed on next page
- **Assignments will be handed in on Gradescope**
 - Link: <https://gradescope.com>
 - Code: D3G5NK
- **All assignments must be turned in BEFORE class on the day that they are due.**
 - This is because we may discuss answers in class and TAs will discuss in section
 - We will accept later homework only in **extraordinary** circumstances. We **may** make arrangements for makeup assignments.
 - Contact the TA immediately if you have a problem or special need
 - Penalty policy for turning in assignments late (subject to change)
 - 1 day late – 10 points
 - 2 days late – 20 points
 - 3 days late – 30 points
 - 4 days late – 40 points
 - 5 days late – not accepted

NOTE: We may change number of assignments. We expect to make a decision on this in first 4-6 weeks of class, but this can change at any time. Basic grading philosophy will remain, but we may change the number of droppable assignments



Graded Homework

- Thought exercises with essay-type answers & analytical problems
- Answers should be neat and legible. I prefer typed answers to the essay questions.
 - If your answer for a question is illegible or too sloppy to allow reasonable grading, then your score for that question will be 0
- Homework is open-book (and open library) but you must work alone or with your designated partner (depending on assignment). You may not consult with other students about the questions or the answers, and you may not consult the answers to previous years' homework and exams. You may use Piazza to obtain clarification about the questions, but you may not use Piazza to discuss answers or approaches to answering the questions. You need to cite any and all external references that you consult. Also, you may not provide assistance to anyone other than your lab partner.
 - You are free to change partners between assignments.
 - Except in unusual circumstances, both partners will receive the same grade for the assignment
 - **NOTE: If one partner essentially abandons the other, I am likely to take this into account in computing the culprit's grade**
- Typically, due about 2-3 weeks after handed out, but I may modify dates



Course Grading formulas

- Throw out grades one of HW#1-3 or one of PA#1-3.
- **Must** hand in HW#4 and PA#4. Do not to cursory job on these. If grade is less than 60, I may throw out the lowest of the other assignments instead.
- I will use optional PA#5 to replace the lowest remaining grade in the formula, except that I will not throw out both PA#1 and PA#2
- Numerical grade formula:
 - $\text{Max}(0.4P+0.6H, 0.6P+0.4H)$
 - H is homework average, P is the programming average
- Letter grade thresholds vary but usually fairly close to 90-80-70



Programming Problems

- Programming projects build on each other
- Typically involve using CIS algorithms discussed in class to determine an “unknown” quantity
- I will provide several debugging data sets with answers and an “unknown” data set
- You should hand in a report containing:
 - Description of the problem and method used to solve it
 - Description of the program structure & who did what
 - The “answer” & short discussion of the answer (why you think it is correct). I recommend that you include also a discussion of the debugging data. Also, the answers should not be embedded somewhere in a program printout. Put them clearly in the report.
 - The documented program listing
 - Include full bibliography and acknowledge any consulting help you get on algorithms (see next page)
- You can use C, C++, Matlab, Python, or something else with concurrence of TA

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Programming Problems (continued)

- For programming, work individually or in teams of two
 - You are **strongly** urged to do this in teams of two
 - The rules for partnering are the same as for the written homework
- Teams should not share code with other teams or assist other teams in debugging but may discuss algorithmic questions
 - But you must cite **ALL** sources, including consulting in your handed-in reports
 - Again, you may not use any material from previous years' assignments, nor should you provide such material to others or post it on public sites.
- If work in a team, grade is identical for the assignment, but I want you both to participate about equally and also tell me who did what.
 - **NOTE:** If one partner essentially abandons the other, I am likely to take this into account in computing the culprit's grade
- Assignments must be handed in on day due. I will accept late homework only in extraordinary circumstances. I may make arrangements for makeup assignments.

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Use of Course Materials

- You are free to make and retain private copies of any course materials for your own use.
- However, you are **not** permitted to share or redistribute any materials (lecture slides, videos, homework assignments, etc.) to any third party.
 - You may share materials with a lab/homework partner who has registered for this course.
 - But that person is also bound by this policy
- Private recording of lectures is prohibited
 - Videos of lectures will be available on Panopto and links to them on Canvas, and their use is subject to the abovementioned policy



A final word about grades

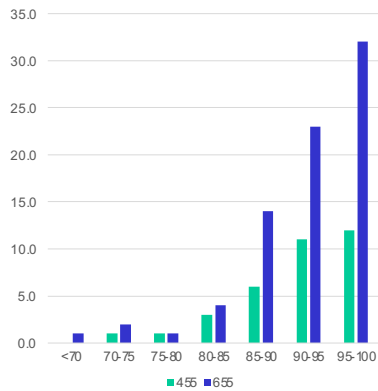
- This course is a lot of work and includes some difficult material. Our experience has been that students who do the work tend to do well, even if they get stuck on one assignment. The grading formula is very forgiving.
- We do not view grades as prizes in a competition so much as a means to help you assess progress and as a motivation to do the work. We give take-home assignments and allow you to work in teams because our experience is that this can help learning, but you both must do the work.
- Do not abuse our trust, either by one member of a team dogging an assignment meant to be done together or (even worse) by breaking the ethics rules. If we find evidence of cheating, we will refer the case to the Ethics Board. The usual consequence of conviction is failure of the course and an annotation on your transcript.



2024 Grades (600.455/655)

	455	655
A+	2	2
A	16	19
A-		6
B+		2
B	3	12
B-	1	
C+		
C	1	1
C-		
F		1
AU		

2025 Numerical
Distribution



Note: The grading formula is very forgiving if students put in the work. It is not always this skewed, but don't panic after the first couple assignments.

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